
CS246: Database Management Systems Lab

Week # 05 (1 Questions, 100 Marks)

Lab session: AL1

Held on: 09-Feb-2026 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 2

Submission dead line: 17:00 hrs, 09-Feb-2026

Submissions accepted from: 16:45 hrs, 09-Feb-2026

Submissions accepted till: 17:30 hrs, 09-Feb-2026

Every one minute delay in submission will result in one mark deduction

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Question 1: (100 points)

Perform the following data definition language related tasks.

Task 01 (0 marks) Create a new database named `week05`. Use the statement `create database week05;`

Task 02 (0 marks) Switch to (use) the database `week05`. Use the statement `use week05;`

Task 03 (48 marks) Create tables with the detailed specification. Refer to `create-table.pdf` manual page

1. (4 marks) Create a table named `table1` with one column: `id` of type `INT`.
2. (4 marks) Create a table named `table2` with one column: `name` of type `CHAR(50)`.
3. (4 marks) Create a table named `table3` with one column: `description` of type `VARCHAR(200)`.
4. (4 marks) Create a table named `table4` with one column: `price` of type `FLOAT`.
5. (4 marks) Create a table named `table5` with one column: `balance` of type `DECIMAL(12,2)`.
6. (4 marks) Create a table named `table6` with one column: `born` of type `DATE`.
7. (4 marks) Create a table named `table7` with one column: `last_login` of type `DATETIME`.
8. (4 marks) Create a table named `table8` with two columns: `roll` (`INT`) and `marks` (`INT`).
9. (4 marks) Create a table named `table9` with two columns: `code` (`CHAR(6)`) and `city` (`CHAR(30)`).
10. (4 marks) Create a table named `table10` with two columns: `emp_id` (`CHAR(8)`) and `name` (`VARCHAR(100)`).
11. (4 marks) Create a table named `table11` with two columns: `sensor_id` (`INT`) and `reading` (`FLOAT`).

12. (4 marks) Create a table named `table12` with five columns having the following data types (in order): `INT`, `CHAR(10)`, `VARCHAR(150)`, `FLOAT`, `DATE`. Choose meaningful column names.

Task 04 (36 marks) Alter table - Modify the tables as per the following description Refer to `alter-table.pdf` manual page

1. (4 marks) Alter `table1` to add a new column `code` of type `CHAR(10)` after the `id` column.
2. (4 marks) Alter `table2` to add a new column `notes` of type `VARCHAR(500)` before the existing `name` column.
3. (4 marks) Alter `table3` to add a new column `weight` of type `FLOAT`.
4. (4 marks) Alter `table4` to add a new column `recorded_at` of type `DATETIME`.
5. (4 marks) Alter `table5` to add the following columns in one `ALTER` statement: `name CHAR(40)`, `comment VARCHAR(300)`, `value FLOAT`, `updated DATETIME`.
6. (4 marks) Alter `table5` to drop the column `name`.
7. (4 marks) Alter `table5` to drop the column `comment` (the `VARCHAR` column).
8. (4 marks) Alter `table5` to change the column `updated` from `DATETIME` to `DATE`.
9. (4 marks) Alter `table5` to rename the column `value` to `amount`.

Task 05 (12 marks) Re-naming table - Re-naming columns Refer to `rename-table.pdf` manual page

1. (4 marks) Rename the table `table5` to `new_table5`.
2. (4 marks) Rename all columns of `new_table5` to `c1`, `c2`, `c3`, `c4`, ... (in the order they currently exist).
3. (4 marks) Rename the table `new_table5` back to `table5` and at the same time rename its columns to: `id_23`, `name_23`, `remark_23`, `val_23`, `dt_23`.

Task 06 (4 marks) Drop the following tables: Refer to `drop-table.pdf` manual page `table1`, `table2`, `table3`, `table4`, `table5`, `table6`, `table7`, `table8`, `table9`, `table10`, `table11`, `table12`.

Task 07 (0 marks) Drop the entire database `week05`. Use the statement `drop database week05;`